

Defence **INS SHIVALIK**

Stealthy PUNCH

The induction of the INS Shivalik is a watershed in indigenous shipbuilding and the first of a new generation of warships

| By Sandeep Unnithan |

For over a decade, the Indian Navy has wrestled with the problem of making its warships stealthier. The solution is embedded in the sleek hull of the INS Shivalik, its newest warship commissioned in Mumbai, the culmination of a decade of design and construction efforts involving thousands of technicians and designers and a final project price tag of Rs 8,000 crore for three vessels.

Modern warships are like floating orchestras. The cacophony of their propellers can be picked up by lurking submarines, their silhouettes, as tall as a five-storeyed building, can be easily detected by radars. This makes them vulnerable to anti-ship missiles and torpedoes. The indigenously built Shivalik is the first to be specifically designed from the drawing board with stealth features that puts India in an exclusive league comprising the US, UK, France, Netherlands, Germany.

Its designers say considerable effort was made in three key areas—the superstructure, heat emissions from its smokestack and noise from its propellers. The ship's sides have been angled at 10 degrees to bounce off radar waves and use radar-absorbent paint. Radar detection of the hull has thus been reduced by up to 90 per cent, allowing the vessel to tip-toe through the high seas. The engines are placed on shock-proof cradles. The engine exhausts have been shrouded and the boat deck and torpedo tubes are concealed behind a curtain.

This stealthy combination means that on a radar, the 5,900-tonne Shivalik would appear to be a warship less than half that size. The analogy is simple: what you cannot detect, you cannot hit. "Shivalik has reduced detection and lock-on ranges for hostile anti-ship missiles and torpedoes," says Rear Admiral K.N. Vaidyanathan, Director General, Naval Design. A quiet predator of the high seas. ■

VITAL STATISTICS Length 143 m Beam 16.9 m

Missile Detection Radar

Barak Anti-Missile System

A total of 32 missiles. Can shoot down incoming missiles that are 10 km away.

AK-630 Gun

Two six-barrelled Gatling guns. Can fire 5,000 rounds per minute. Range 4 km.

Helicopters

Two Sea King twin-engined anti-submarine or transport helicopters

COVERT FACTOR

Acoustic Signature>>

Propeller

'Delayed cavitation inception' design is less noisy, leaves narrow wake



Engine



Two US-made GE LM2500 gas turbines on vibration-proof mounts to reduce vibration

Infra Red Signature>>



A special educator diffuser device cools and diffuses the smoke plume rising from the exhaust to reduce the ship's thermal image. Hot metal temperatures are also reduced.